

AUTOMATED PRICE INTELLIGENCE

High-Frequency Inflation Nowcasting via Automated Price Pipeline

Turning online product prices into high-frequency CPI signals and real-time economic intelligence.



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Introduction: What is CPI & Why Does It Matter?

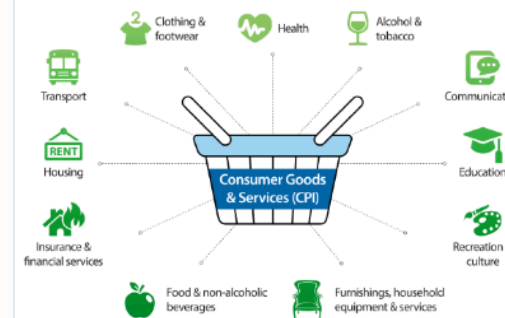
Consumer Price Index (CPI) Defined

A macroeconomic indicator that tracks the aggregate price of a fixed monthly "**shopping basket**" of consumer goods and services typically purchased by households across **12 official COICOP categories**.

Why CPI Matters in the Real World:

- **Monetary Policy:** Guides central bank interest rates and currency defense.
- **Purchasing Power & Cost of Living:** Sets wage negotiations, pension adjustments, and public subsidies.
- **Planning & Real Contracts:** Indexes commercial leases, government budgets, and supply chain agreements.

What Is a Consumer Price Index?



Picture one shopping basket of good and services holding everything a typical family buys in a month.

- A fixed list of hundreds of everyday goods & services
- Re-priced every month in the same shops
- Combined into **one number you can track** over time known as CPI

Problem Statement: Limitations of Traditional CPI

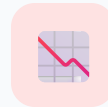
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Severe Time-Lag

Official CPI statistics are published monthly or quarterly with a **2 to 4-week delay**. Decisions are made using outdated economic snapshots.

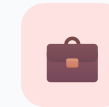
- Delayed response to sudden price shocks
- Slow detection of local commodity spikes



Low Data Frequency

Prices are collected once a month at discrete survey intervals, missing high-volatility shifts in essential items like fuel, meat, and fresh food.

- Lack of daily or weekly visibility
- No high-frequency intra-month trend indicators



High Operational Friction

Traditional CPI relies on labor-intensive manual price audits at brick-and-mortar stalls, which is expensive, difficult to scale, and prone to collection bias.

- High surveying and field cost
- Limited geographic and store coverage

Project Objectives: Real-Time Inflation Intelligence



1. Automated Ingestion

Build a robust, automated daily data harvesting pipeline extracting live product prices, units, and categories across modern supermarkets and e-commerce outlets.



2. COICOP Harmonization

Standardize scraped products, normalize units (weight/volume), and map retail goods to international **COICOP classifications** to compute high-frequency price indices.



3. Nowcasting & Validation

Deploy machine learning and time-series models to **nowcast the official CPI** weeks ahead of release, evaluating tracking error and residual bias.

Core Outcome

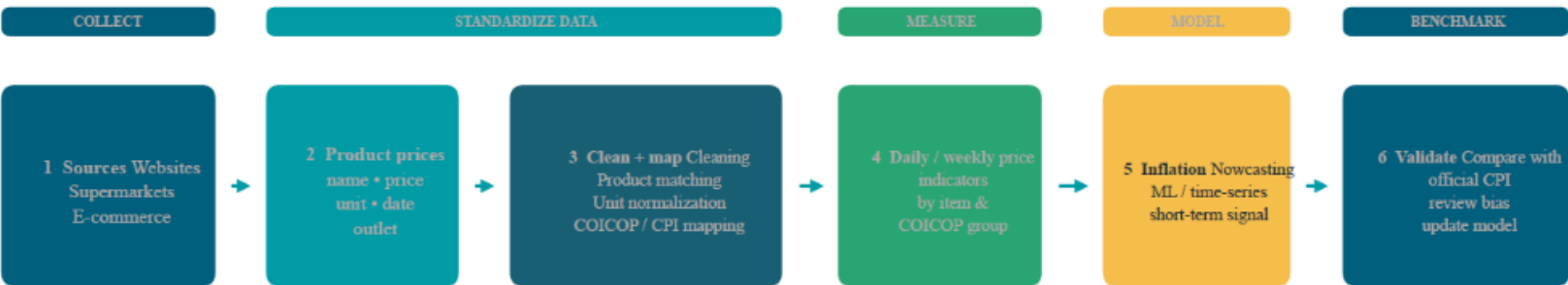
Empowering economists, policymakers, and business leaders with leading indicators rather than trailing post-mortems.

DAILY HIGH-FREQUENCY SIGNALS

Methodology: Web-Scraped Prices for Nowcasting

Framework: Web-Scraped Prices for Inflation Nowcasting

Turn online product prices into high-frequency CPI signals, then validate them against official CPI.



STEP 1

Collect

Supermarket websites, e-commerce, and APIs. Extract product name, price, unit, date, and outlet.

STEP 2

Standardize

Data cleaning, product matching, unit normalization, and mapping into the 12 COICOP groups.

STEP 3

Measure

Calculate daily & weekly high-frequency price indices by individual item and group level.

STEP 4

Model

Machine Learning and time-series nowcasting models generate leading inflation signals.

STEP 5

Benchmark

Validate nowcast accuracy against official monthly government CPI releases; optimize model weights.

Methodology: End-to-End Data Pipeline Architecture

Orchestration & Infrastructure

Apache Airflow schedules ingestion and transformation DAGs within a containerized **Docker Compose** stack.

PostgreSQL Medallion Architecture

- **Bronze (Raw)**: Stores raw scraped payloads, API records, and CSVs.
- **Silver (Cleaned – dbt)**: Deduplicates, cleans, normalizes units, and staging models.
- **Gold (Marts – dbt)**: Curated star schema tables with dimension and fact models.

Serving & Analytics Plane

Metabase powers interactive BI dashboards tracking real-time inflation trends and price volatility across sectors.

